Body Mass Index

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The body mass index (BMI) is a calculation used to quantify the severity of obesity. It utilizes the patient's height and weight to arrive at a value used by clinicians to diagnose obesity. A patient with a BMI ≥ 24 is defined as overweight. A BMI ≥ 30 indicates obesity. A BMI ≥ 40 indicates severe (previously referred to as morbid) obesity. Some clinicians utilize the term super-obesity to refer to patients with a BMI ≥ 50. This is a categorization we need to remember.

Patient Classification	**BMI**
Normal	<25
Overweight	25-29.9
Moderate Obesity	30-34.9
Severe Obesity	35-39.9
Morbid Obesity	≥40

The BMI is calculated as follows:

BMI = (Pt's weight in Kg)/(Pt's height in meters squared)

Let's try an example:

What is a patient's degree of obesity if he is 1.75 meters tall and weighs 145 Kg?

To classify the patient according to the scale we described above, we need to divide the weight in kilograms (145) by the height in meters (1.75) squared. So,

BMI = (145 Kg)/(1.75)(1.75)

Because this patient's BMI is 47, we would classify him as morbidly obese.

By itself, the calculation is fairly simple. It becomes slightly more complicated when the patient's values for height and weight are given in US Standard terms instead of metric. For example:

What is a patient's BMI who weighs 249 lbs. and is 5'6" tall?

This problem first requires that we perform two unit conversions prior to calculating our answer. To begin with, we need to convert pounds into kilograms. This unit conversion is expressed as

1 pound = 0.45 kilograms

OR

1 kilogram = 2.2 pounds

By remembering this conversion, we can easily convert pounds into kilograms by dividing it by 2.2. Setting it up using factor analysis, it looks like this:

$$\frac{249 \text{ pounds}}{1} \times \frac{1 \text{ kilogram}}{2.2 \text{ pounds}} = 113.2 \text{ Kg}$$

Now that we've obtained the numerator of our formula, we need to determine the height in meters. It is easiest to first convert the height into inches and then into meters.

5 feet 6 inches = (5 X 12 inches) + 6 inches

5 feet 6 inches = (60 inches) + 6 inches

5 feet 6 inches = 66 inches

If we remember that 1 inch = 2.54 centimeters and there are 100 centimeters in a meter, we can use factor analysis to set up our conversion from inches into meters.

$$\frac{66 \text{ inches}}{1} \times \frac{2.54 \text{ cm}}{1 \text{ inch}} \times \frac{1 \text{ meter}}{100 \text{ cm}} = 1.7 \text{ m}$$

Now we have the factors we need to calculate the BMI.

$$BMI = \frac{113.2\ Kg}{1.7\ m^2} = \frac{113.2\ Kg}{(1.7m)(1.7m)} = 39.2 \text{ or } 39\ Kg/m^2$$

This patient's BMI is 39, which categorizes him as severely obese.
We'll do one more as an example:

Is a patient who weighs 199 pounds and is 5 feet 10 inches tall obese?

Convert the patient's weight from pounds into kilograms using factor analysis.

$$\frac{199\ \text{pounds}}{1} \times \frac{1\ \text{kilogram}}{2.2\ \text{pounds}} = 90.5\ Kg$$

Convert the patient's height into inches:

5 feet 10 inches = (5 X 12) + 10

5 feet 10 inches = 70 inches

Now, use factor analysis to convert inches into centimeters and then centimeters into meters.

$$\frac{70\ \text{inches}}{1} \times \frac{2.54\ \text{cm}}{1\ \text{inch}} \times \frac{1\ \text{meter}}{100\ \text{cm}} = 1.78\ m$$

Now we have the factors we need to calculate the BMI.

$$BMI = \frac{90.5\ Kg}{1.78\ m^2} = \frac{90.5\ Kg}{(1.78m)(1.78m)} = 28.5 \text{ or } 29\ Kg/m^2$$

Because our definition of obesity begins at BMI of 30, we recognize that this patient is overweight, but not obese.

BMI Practice Problems

Calculate the BMI for each of the following patients:
(round answers up to nearest whole number)

1. A patient who is 1.8 meters tall and weighs 94 Kg.

2. A patient who is 1.51 meters tall and weighs 120 Kg.

3. A patient who is 1.7 meters tall and weighs 82 Kg.

4. A patient who is 1.63 meters tall and weighs 96 Kg.

5. A patient who is 1.84 meters tall and weighs 75 Kg.

6. A patient who is 1.63 meters tall and weighs 110 Kg.

7. A patient who is 1.72 meters tall and weighs 200 Kg.

8. A patient who is 1.9 meters tall and weighs 140 Kg.

9. A patient who is 1.59 meters tall and weighs 133 Kg.

10. A patient who is 1.6 meters tall and weighs 122 Kg.

11. A patient who is 1.77 meters tall and weighs 145 Kg.

12. A patient who is 1.65 meters tall and weighs 87 Kg.

13. A patient who is 1.48 meters tall and weighs 104 Kg.

14. A patient who is 1.91 meters tall and weighs 115 Kg.

15. A patient who is 1.68 meters tall and weighs 76 Kg.

16. A patient who is 1.75 meters tall and weighs 152 Kg.

17. A patient who is 1.54 meters tall and weighs 108 Kg.

18. A patient who is 1.58 meters tall and weighs 138 Kg.

19. A patient who is 1.88 meters tall and weighs 92 Kg.

20. A patient who is 1.82 meters tall and weighs 185 Kg.

21. A patient who is 1.59 meters tall and weighs 72 Kg.

22. A patient who is 1.78 meters tall and weighs 201 Kg.

23. A patient who is 1.89 meters tall and weighs 168 Kg.

24. A patient who is 1.94 meters tall and weighs 115 Kg.

25. A patient who is 1.91 meters tall and weighs 163 Kg.

26. A patient who is 5 feet 7 inches tall and weighs 312 pounds.

27. A patient who is 4 feet 11 inches tall and weighs 172 pounds.

28. A patient who is 6 feet 0 inches tall and weighs 159 pounds.

29. A patient who is 6 feet 2 inches tall and weighs 248 pounds.

30. A patient who is 5 feet 6 inches tall and weighs 196 pounds.

31. A patient who is 5 feet 4 inches tall and weighs 273 pounds.

32. A patient who is 5 feet 11 inches tall and weighs 302 pounds.

33. A patient who is 5 feet 1 inches tall and weighs 124 pounds.

34. A patient who is 5 feet 3 inches tall and weighs 169 pounds.

35. A patient who is 5 feet 5 inches tall and weighs 184 pounds.

36. A patient who is 6 feet 3 inches tall and weighs 258 pounds.

37. A patient who is 5 feet 9 inches tall and weighs 182 pounds.

38. A patient who is 6 feet 4 inches tall and weighs 415 pounds.

39. A patient who is 5 feet 1 inches tall and weighs 123 pounds.

40. A patient who is 5 feet 9 inches tall and weighs 201 pounds.

41. A patient who is 5 feet 10 inches tall and weighs 175 pounds.

42. A patient who is 6 feet 0 inches tall and weighs 190 pounds.

43. A patient who is 5 feet 8 inches tall and weighs 263 pounds.

44. A patient who is 5 feet 6 inches tall and weighs 221 pounds.

45. A patient who is 5 feet 11 inches tall and weighs 401 pounds.

46. A patient who is 5 feet 0 inches tall and weighs 201 pounds.

47. A patient who is 5 feet 5 inches tall and weighs 280 pounds.

48. A patient who is 6 feet 1 inches tall and weighs 320 pounds.

49. A patient who is 5 feet 10 inches tall and weighs 260 pounds.

50. A patient who is 5 feet 9 inches tall and weighs 318 pounds.

BMI Answers

BMI		Height in meters	Wt in Kg	BMI
1. 29	26.	1.7	141.8	49
2. 53	27.	1.5	78.2	35
3. 28	28.	1.83	72.3	22
4. 36	29.	1.88	112.7	32
5. 22	30.	1.68	89.1	32
6. 41	31.	1.63	124.1	47
7. 68	32.	1.8	137.3	42
8. 39	33.	1.55	56.4	23
9. 53	34.	1.6	76.8	30
10. 48	35.	1.65	83.6	31
11. 46	36.	1.91	117.3	32
12. 32	37.	1.75	82.7	27
13. 47	38.	1.93	188.6	51
14. 32	39.	1.55	55.9	23
15. 27	40.	1.75	91.4	29
16. 50	41.	1.78	79.5	25
17. 46	42.	1.83	86.4	26
18. 55	43.	1.73	119.5	40
19. 26	44.	1.68	100.5	36
20. 56	45.	1.8	182.3	56
21. 28	46.	1.52	91.4	40
22. 63	47.	1.65	127.3	47
23. 47	48.	1.85	145.5	43
24. 31	49.	1.78	118.2	37
25. 45	50.	1.75	144.5	47